

CS 222: Computer Organization and Architecture

MSE Max Marks: 50, Max Time: 120 minutes

(Answer precisely and mention explicitly all the steps)

1. Compute the PRODUCT of the two IEEE single-precision FP numbers (BFD00000, BFB00000) using FP multiplication algorithm, and express the result as a decimal number. [5]
2. Assume an 8-bit floating point representation based on the IEEE floating point format as follows. [12]
 - There is a sign bit in the most significant bit.
 - The next 3 bits are the exponent. The exponent bias is $2^3 - 1 - 1 = 3$.
 - The last 4 bits are the fraction.
 - The representation encodes numbers of the form: $V = (-1)^s \times M \times 2^E$, where M is the significand and E is the biased exponent.

The rules are same as that discussed in class the IEEE standard (normalized, denormalized, representation of 0, infinity, and NAN). Answer the followings

 - (i) Represent -0 (minus Zero) in binary form. Compute the values for significand and exponent.
 - (ii) What is the decimal value of smallest denormalized (negative) number?
 - (iii) What is the decimal value of largest normalized (positive) number?
3. Consider a 32-bit microprocessor whose bus cycle is the same duration as that of a 16-bit microprocessor. Assume that, on average, 20% of the data items are 32 bits long, 40% are 16 bits long, and 40% are only 8 bits long. Calculate the improvement achieved when access the data items with the 32-bit microprocessor compared with 16-bit processor. (Note that the data items are not continuously allocated). [4]
4. Explain the terms temporal and spatial locality using a suitable C-code segment. [4]
5. A two-way set-associative cache has lines of 16 bytes and a total size of 8 kbytes. The 64-Mbyte main memory is byte addressable. [8]
 - (a) Show the format of main memory addresses.
 - (b) For the hexadecimal main memory addresses 111111, 666666, BBBB, show the values of Tag, Set, and Word fields.
6. Give a high-level chip diagram of how to construct an 8-Mbyte computer memory using eight 1-Mbyte chips. Be sure to show the address lines in your diagram and what the address lines are used for. [4]
7. Discuss how the data bits are stored using FM encoding mechanism in a magnetic disk. What is the advantage of this mechanism. [4]
8. Consider a single-platter disk with the following parameters: rotation speed: 7200 rpm; number of tracks on one side of platter: 30,000; number of sectors per track: 600; seek time: one ms for every hundred tracks traversed. Let the disk receive a request to access a random sector on a random track and assume the disk head starts at track 0. [4]
 - a. What is the average seek time?
 - b. What is the average rotational latency?
9. Assume an instruction set that uses a fixed 16-bit instruction length. Operand specifiers are 6 bits in length. There are 12 two-operand instructions and 128 zero-operand instructions. What is the maximum number of one-operand instructions that can be supported? [5]